



Republic of Rwanda
Ministry of Education



SWDIA502

INDUSTRIAL ATTACHMENT PROGRAM (IAP)

Integrate the workplace

Competence

RQF Level: 5

Learning Hours 200

Credits: 20



Sector: ICT AND MULTIMEDIA

Trade: SOFTWARE DEVELOPMENT

Module Type: Specific

Curriculum: CBSIND5001-TVET CERTIFICATE V IN SOFTWARE DEVELOPMENT

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Purpose statement	This module describes the skills, knowledge and attitudes required to integrate learners/trainees in workplace for an industrial attachment program to gain work-based experience for future employment. The module will enable the learner/trainees to Demonstrate supervisory skills, enhance innovation during IAP and to promote professional skills transfer.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			N.A

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Demonstrate supervisory skills	1.1. Labour contracts are correctly developed based on labour market law
	1.2. Supervisor roles and responsibilities are properly fulfilled based on organisation structure.
	1.3. Supervisor qualities and skills are accurately exhibited at the workplace.
	1.4. Document related to the work are appropriately filed.
2. Enhance creativity and innovation during IAP program	2.1. Innovation concepts are properly explained based on industrial attachment program core values.
	2.2. Innovation strategies during industrial attachment program are properly described based on assigned industry duties.
	2.3. Innovation process is effectively integrated in line with industrial attachment program program.
3. Promote and implement professional skills transfer	3.1. Implementation plan of competencies acquired during industrial attachment program is properly elaborated in accordance with labour market needs.
	3.2. Knowledge, skills and attitudes acquired during industrial attachment program are effectively applied based on developed implementation plan.
	3.3. Industrial attachment program report presentation is properly performed.

Learning outcomes	At the end of the module the learner will be able to: 1. Demonstrate supervisory skills 2. Enhance creativity and innovation during industrial attachment program 3. Promote professional skills transfer
Learning outcome 1: Demonstrate supervisory skills	Learning hours: 15
Indicative content	
• Identification of labour contract ✓ Definition of key terms ✚ Labor ✚ Contract ✚ Labor market ✓ Main components of labor contract ✚ Client ✚ Contractor ✚ duration ✓ Types of labor contract ✚ permanent labor contracts ✚ fixed-term contracts ✚ casual labor contracts ✓ Low governing contract in Rwanda • Identification of supervisor 's roles and responsibilities ✓ Supervisor responsibilities ✚ Workflow management ✚ Training new employees/intern ✚ Creating team schedule ✚ Evaluating employee/interns performance ✓ Supervisor's roles • Supervisor qualities and skills ✓ Meaning of key terms ✚ Skills ✚ Knowledge ✚ Qualities ✓ Supervisor skills ✚ Time management ✚ Communication skills ✚ Critical thinking ✓ Supervisor qualities ✚ Managing workflow	

- ✚ Training new hires
- ✚ Creating and managing team schedules
- ✚ Reporting
- ✚ Evaluating performance and providing feed back
- ✚ Help in employees conflict resolution

- **Filing work related documents**

- ✓ Definition of key terms
 - ✚ Filing
 - ✚ File
 - ✚ document
- ✓ Different ways to file documents in a filing cabinet
 - ✚ Alphabetical
 - ✚ Category of Date
 - ✚ Numerical
 - ✚ Combination

Resources required for the learning outcome

Equipment	• Computer ,projector ,chalk board ,Flipchart stand
Materials	• Markers, chalks, pen, Scotch tape, Papers, flip chat
Tools	• Labor contract sample • Visual aids
Facilitation techniques	• Demonstration and simulation • Individual research • Group discussion and presentation • Learning through interaction of guest speakers(Practitioners) • Learning through visits on the relevant institutions • Case study
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment

Indicative content

- **Innovation and creativity concepts**
 - ✓ Definition of key terms
 - ✚ Creativity
 - ✚ Innovation
 - ✓ Types of innovation
 - ✚ Radical
 - ✚ Incremental
 - ✚ Disruptive
 - ✚ Architecture
 - ✓ Classification of innovation
 - ✚ Product innovation
 - ✚ Process innovation
 - ✚ Marketing innovation
 - ✚ Organizational innovation
 - ✓ Technological innovation processes
 - ✚ Product oriented conversion process
 - ✚ Technology/market oriented conversion process
 - ✓ The holistic TVET innovation pattern
 - ✚ Political
 - ✚ Practical
 - ✚ Scientific
 - ✓ Types of creativity
 - ✚ Deliberate and cognitive
 - ✚ Deliberate and emotional
 - ✚ Spontaneous and cognitive
 - ✚ Spontaneous and emotional
 - ✓ Core values related to IAP
 - ✚ Competences acquired
 - ✚ Innovation process
 - ✚ Skills transfer
- **Description of innovation and creativity strategies during IAP.**
 - ✓ Types of innovation strategies
 - ✚ Proactive
 - ✚ Active
 - ✚ Reactive
 - ✚ Passive
 - ✓ **Examples of Innovation strategies**

-  Exploration
-  Cooperation
-  Entrepreneurship

- **Integration of innovation and creativity process**

- ✓ Definition of integration of innovation
- ✓ Purpose of integration of innovation
 -  Introducing novelty
 -  Increase market share
 -  Increase revenue
 -  Increase customer satisfaction
- ✓ Integration of Innovation process
 -  Idea Generation and Mobilization
 -  Advocacy and Screening
 -  Experimentation
 -  Commercialization
- ✓ Creativity process
 -  Preparation /saturation
 -  Incubation
 -  Elimination and verification/implementation

Resources required for the indicative content

Equipment	• Computer, Projector, white/chalkboard, flipchart stand
Materials	• Flip Chart, markers/chalks, paper,scotch tape,pen,pencil,stickers
Tools	• Reference books, hand out notes,visual aids
Facilitation techniques	• Demonstration and simulation • Individual research • Trainer guided • Group discussion and presentation • Four step method • Project method • Technical discussion
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment

Indicative content

- **Elaboration of the implementation plan of competences acquired during IAP**
 - ✓ Definition of implementation plan
 - ✓ Description of components of implementation plan
 - ✚ Task to be performed
 - ✚ Steps required
 - ✚ Resource's needed
 - ✚ Risks identification
 - ✚ Schedules
 - ✚ Task Verification
- **Application of knowledge, skills and attitudes acquired during IAP implementation plan Execution**
 - ✓ Definition execution of implementation plan?
 - ✓ Performance evaluation of the task carried out
 - ✚ Define the learning objective
 - ✚ Identifies concepts or skills students need to demonstrate
 - ✚ Identify the level of performance and their points of values
 - ✚ Identify criteria for each level of performance
 - ✚ Create grading system based on possible points earned
 - ✓ Preparation of report on the work done
 - ✚ Elements of report writing
 - ✚ The different types of reports
 - ✓ IAP report presentation
 - ✚ IAP report presentation definition
 - ✚ How to write IAP report presentation
 - ✚ Steps used for presenting AIP report

Resources required for the indicative content

Equipment	• Equipment available at workplace, flipchart stand
Materials	• Raw /consumable materials at the workplace • Markers/Chalks,papers,stickers,pens,pencil,
Tools	• Vocational tools available at the workplace • Checklists for impact assessment,sample IAP implementation plan ,logbooks,
Facilitation techniques	• Individual and group work • Individualized • Trainer guided • Technical discussion and presentation • Project method
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment

References:

1. Bing Wen SHAN BIAN ZHU, Cement Concrete Dodgson, Mark, Gann, David and Salter, Ammon. 2008. *The Management of Technological Innovation: Strategy and Practice. Completely rev. and updated.* Oxford: Oxford University Press
2. AFL-CIO. "Training and Apprenticeships." <http://www.aflcio.org/Learn-About-Unions/Training-and-Apprenticeships>.
3. Central Intelligence Agency. "The World Factbook." <https://www.cia.gov/library/publications/the-world-factbook/index.html>.
4. Clark, John Bates. *Essentials of Economic Theory: As Applied to Modern Problems of Industry and Public Policy.* New York: A. M. Kelley, 1907, 501